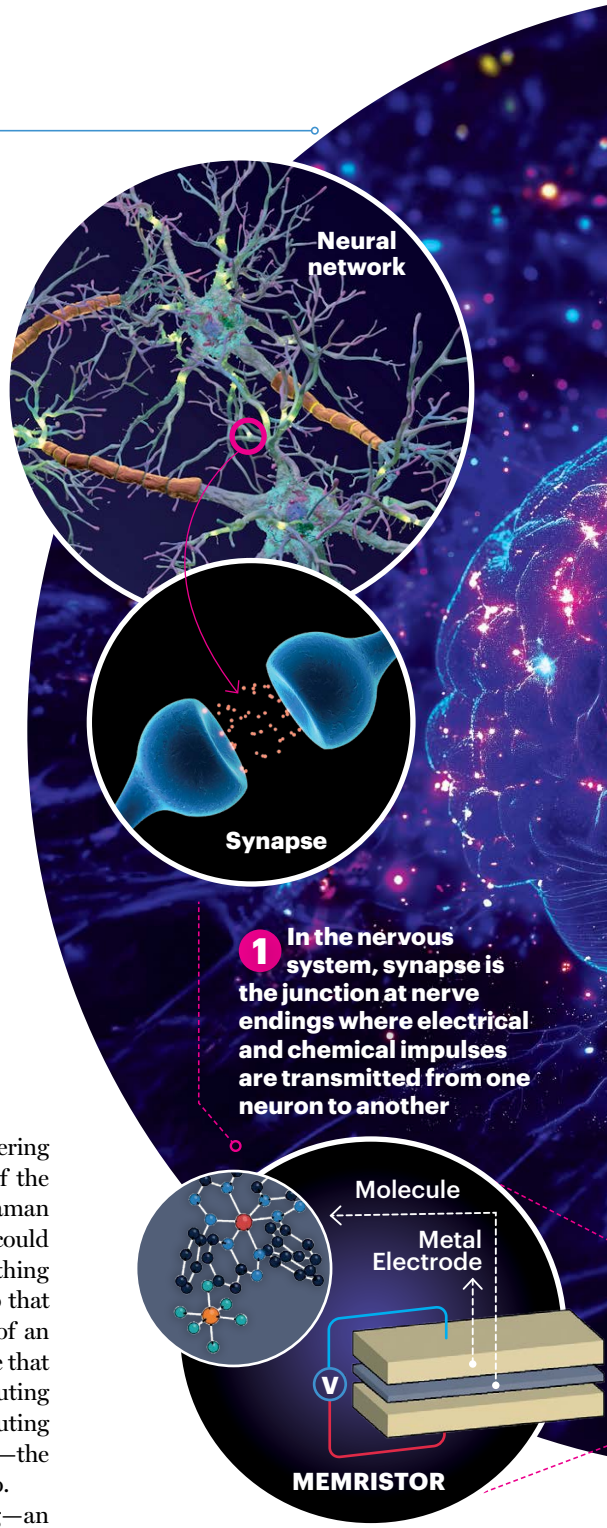
THE HOLY GRAIL OF COMPUTING

IN A GIANT LEAP TOWARDS BRAIN-INSPIRED COMPUTING, IISc RESEARCHERS HAVE ENGINEERED A DEVICE THAT PROCESSES INFORMATION LIKE NEURAL SYNAPSES, AND PROMISES A LEAP IN SPEED, ENERGY EFFICIENCY AND DATA SECURITY

By **Ajay Sukumaran**



In a corner of the Centre for Nano Science and Engineering (CeNSE) building inside the serene, leafy environs of the Indian Institute of Science (IISc), Bengaluru, is a Raman spectroscopy lab—the most silent working space you could ever imagine. Here, acoustic noise is reduced to something lower than a whisper, that is, less than 30 decibels, so that sensitive measurements, like probing the properties of an individual molecule, can be carried out. It's in this place that a breakthrough, one that promises to place India on the global map of computing inventions, was achieved recently. The invention was of a new device—a computing accelerator—that processes data not in the conventional way a microprocessor—the brain of a computer—does. But closer to how an actual human brain would do.

Welcome to the world of neuromorphic, or brain-inspired, computing—an interdisciplinary field that tries to find a meeting point between neuroscience and computer engineering. One of its biggest sources of inspiration is the synapse, the junction located at nerve endings inside a human body where electrical impulses are transmitted between two neurons or between a neuron and a muscle cell. "If you look at the synapse, it can store data in thousands of states between one axon (which transmits a neuron signal) and a dendron (which receives it). So, the question is, if the brain can store data in so many states, why can't electronic devices," says Sreetosh Goswami, principal investigator of a seven-member IISc team that carried out research and development of the accelerator. Their research paper was published in the British weekly scientific journal *Nature* on September 11.

At the heart of this accelerator is a bit of chemistry—the development of a type of film whose molecules, when excited by an electric pulse, can create various unique states of conductance or ease of data flow. The exact number is 16,500 different states. Compare this with only two conductance states in a digital computer. This is because the molecular film encodes the electric signal in a continuum of molecular states unlike a digital code that breaks it down to a binary